

Project Development Phase

preprocessing steps & business questions with visualization

Date	18 July 2026
Team ID	SWTID-2026-2426
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits

1. Data Preprocessing Steps

The food ordering dataset used for this analysis includes fields such as Order Id, User Id, Age Group, City, Company, Cuisine, Meal Type, Restaurant Type, Order Time, Time Taken To Order, Delivery Fee, Discount Applied, Hunger Level, Mood, Rainy Weather, Is Repeat Order and Number Of Records. The following steps were carried out on this data before building the Tableau dashboards.

Step 1: Data Collection

The raw food ordering dataset was imported into Tableau Desktop as the primary data source, bringing in all order-level attributes (Order Id, Age Group, City, Cuisine, Meal Type, Company, Restaurant Type, Delivery Fee, Order Time, etc.) needed for downstream analysis.
Imported the food ordering dataset into Tableau.

Step 2: Data Cleaning

Each field was checked for gaps and inconsistencies before analysis. Rows with missing values in key columns (e.g. Delivery Fee, Time Taken To Order, Rating Given) were identified, duplicate order records were removed so orders were not double-counted, and incorrect entries (e.g. mislabeled City or Cuisine values) were corrected to keep the dataset reliable.

- Checked for missing values.
- Removed duplicate records.
- Corrected incorrect data entries.

Step 3: Data Transformation

Columns were converted into the appropriate data types (Order Value and Delivery Fee as numeric, Order Time as a date/time field, Age Group and Meal Type as categorical dimensions) and renamed for clarity. Calculated fields such as Total Orders (a count of Order Id) and Average Rating (an average of Rating Given) were created to directly support the business questions being analysed.

- Converted columns into appropriate data types.
- Renamed columns for better understanding.
- Created calculated fields such as Total Orders and Average Rating.

Step 4: Data Validation

The cleaned and transformed dataset was checked against the original source to confirm record counts matched, and filters on City, Cuisine, Meal Type and Day Type were applied and tested to make sure segment-level totals stayed consistent. Once validated, the dataset was finalized as the basis for the dashboards covering all 10 business questions.

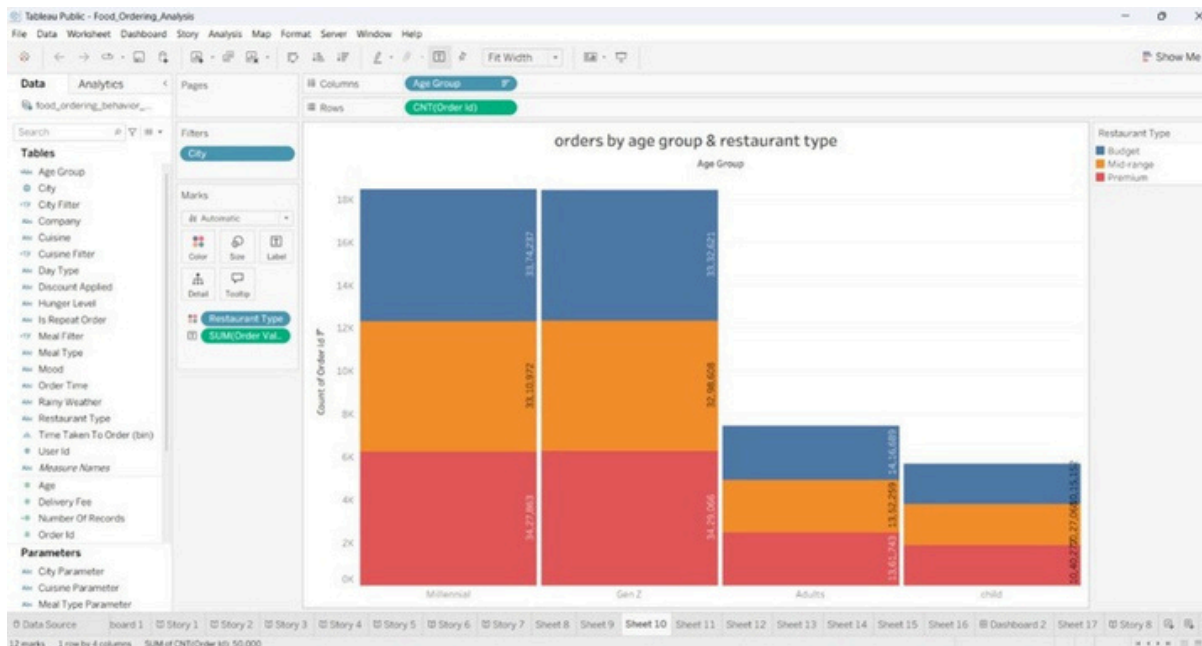
- Verified data accuracy.
- Applied filters (City, Cuisine, Meal Type, Day Type, etc.).
- Prepared the cleaned dataset for dashboard creation.

2. Business Questions with Visualization

Business Question 1: Which age group places the highest number of food orders?

Observation: The dashboard identifies the age group with the highest order volume and their preferred restaurant type.

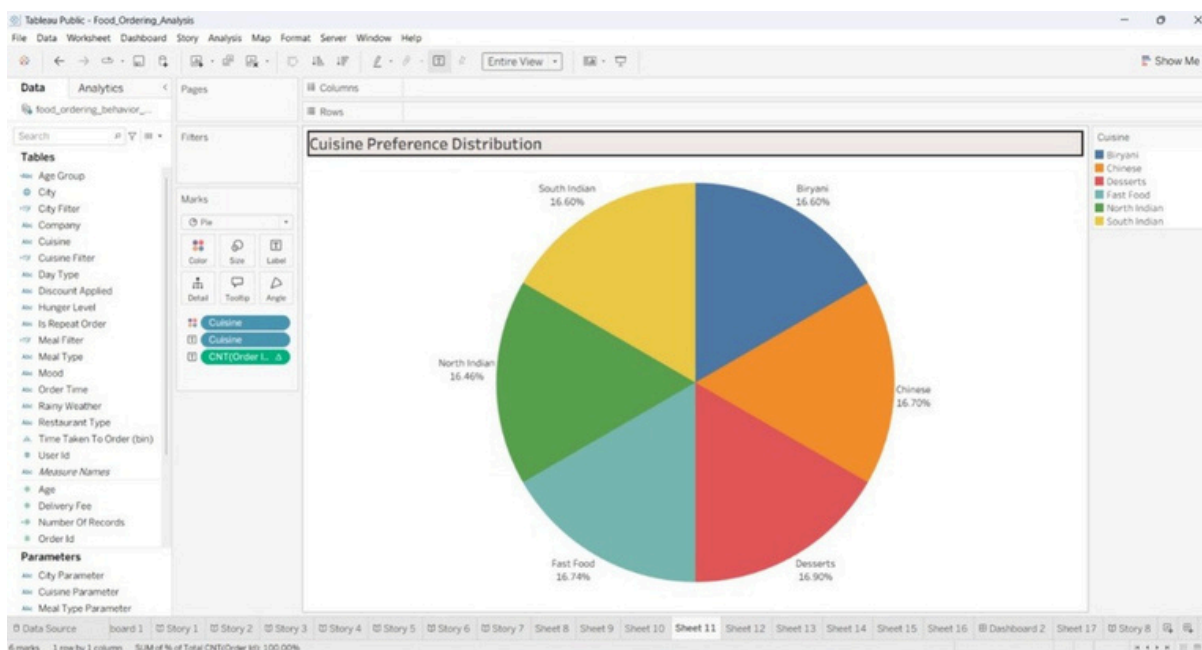
Visualization:



Business Question 2: How are customer cuisine preferences distributed?

Observation: The pie chart shows the percentage share of each cuisine and highlights the most preferred cuisine.

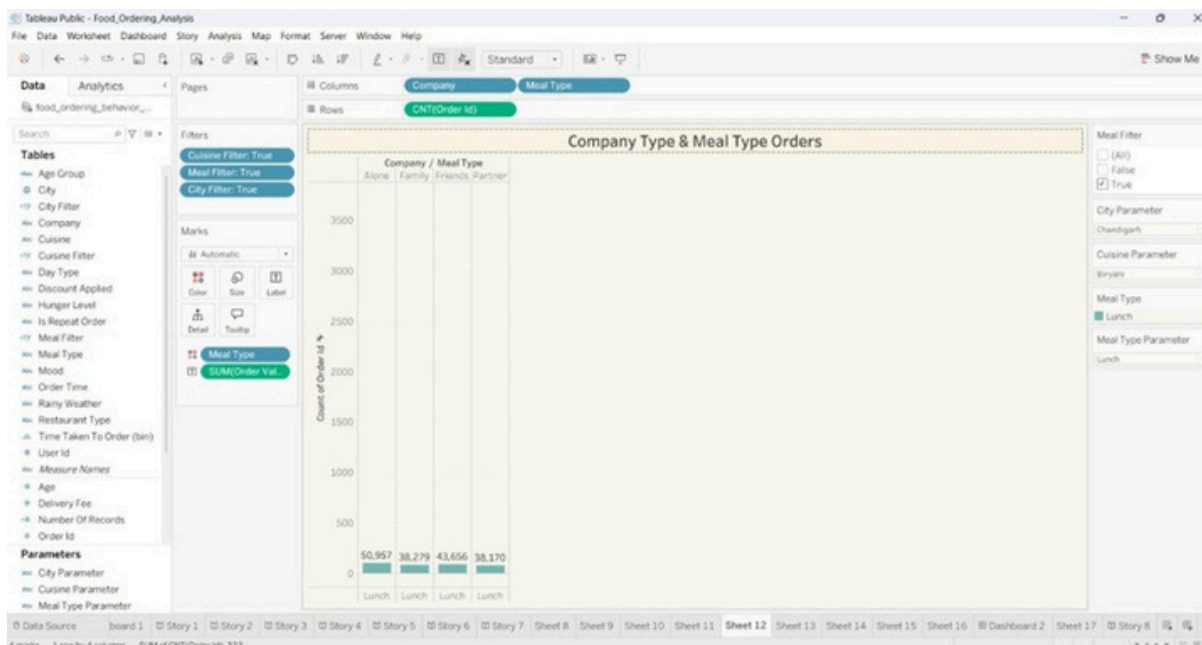
Visualization:



Business Question 3: Which company type receives the highest orders for each meal type?

Observation: The dashboard compares meal type orders across different company types.

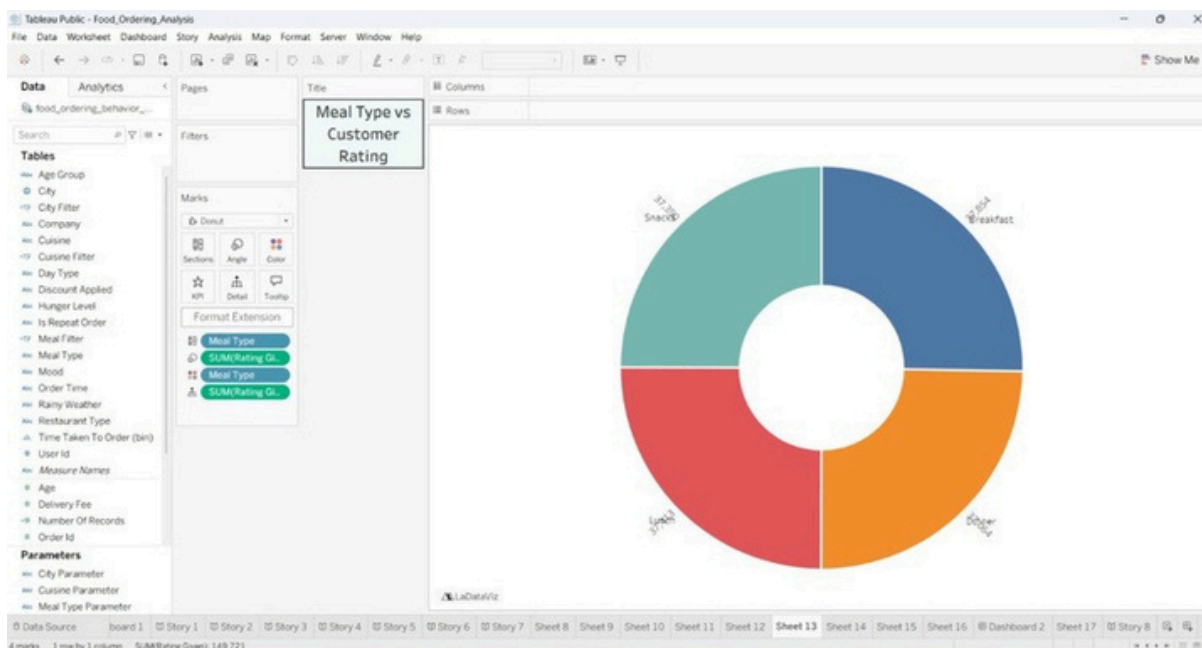
Visualization:



Business Question 4: How do customer ratings vary across meal types?

Observation: The donut chart shows customer ratings for each meal type.

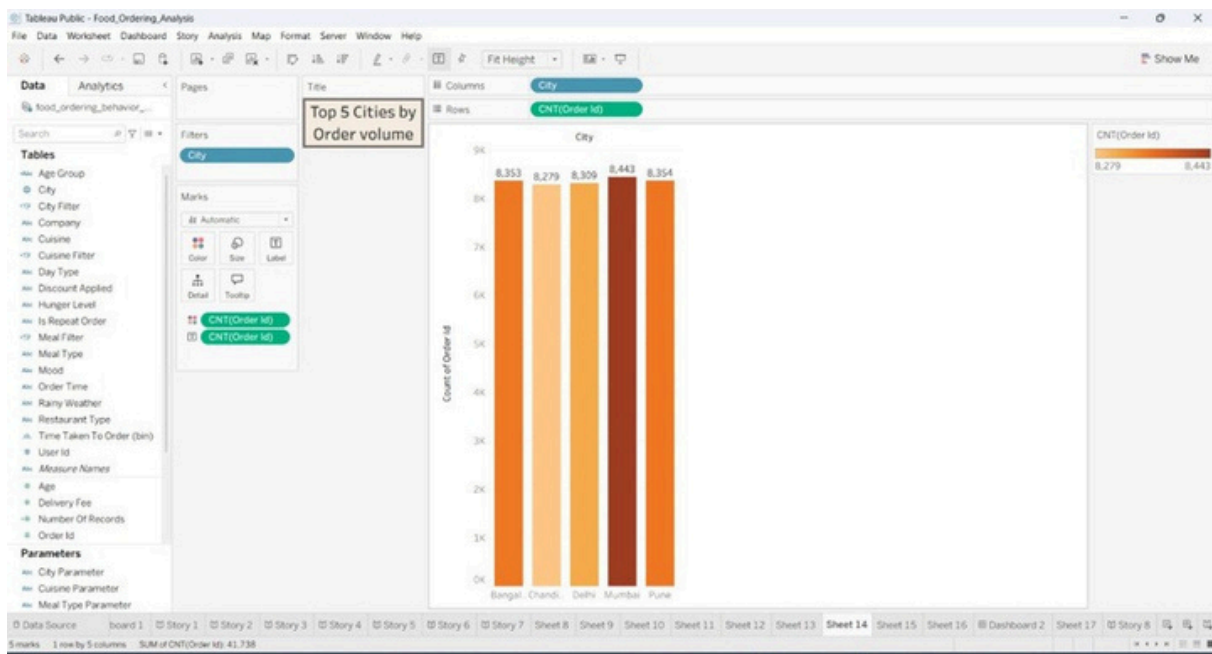
Visualization:



Business Question 5: Which are the top five cities by order volume?

Observation: The dashboard highlights the top five cities with the highest number of food orders.

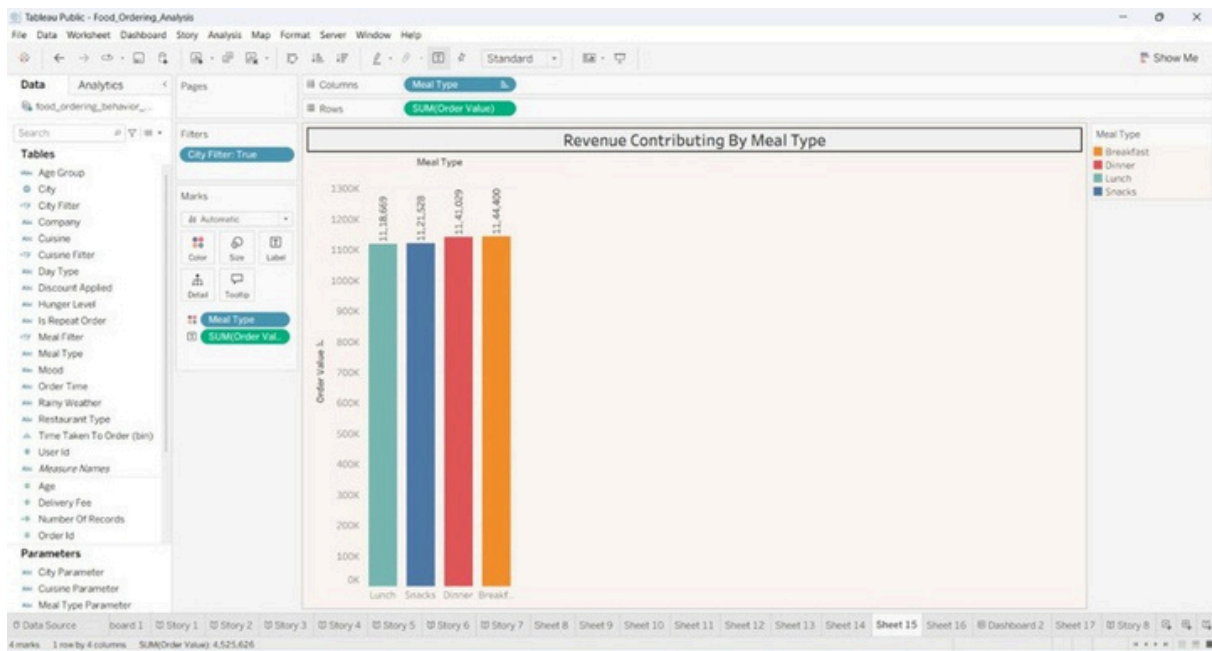
Visualization:



Business Question 6: Which meal type contributes the most revenue?

Observation: The dashboard identifies the meal type generating the highest revenue.

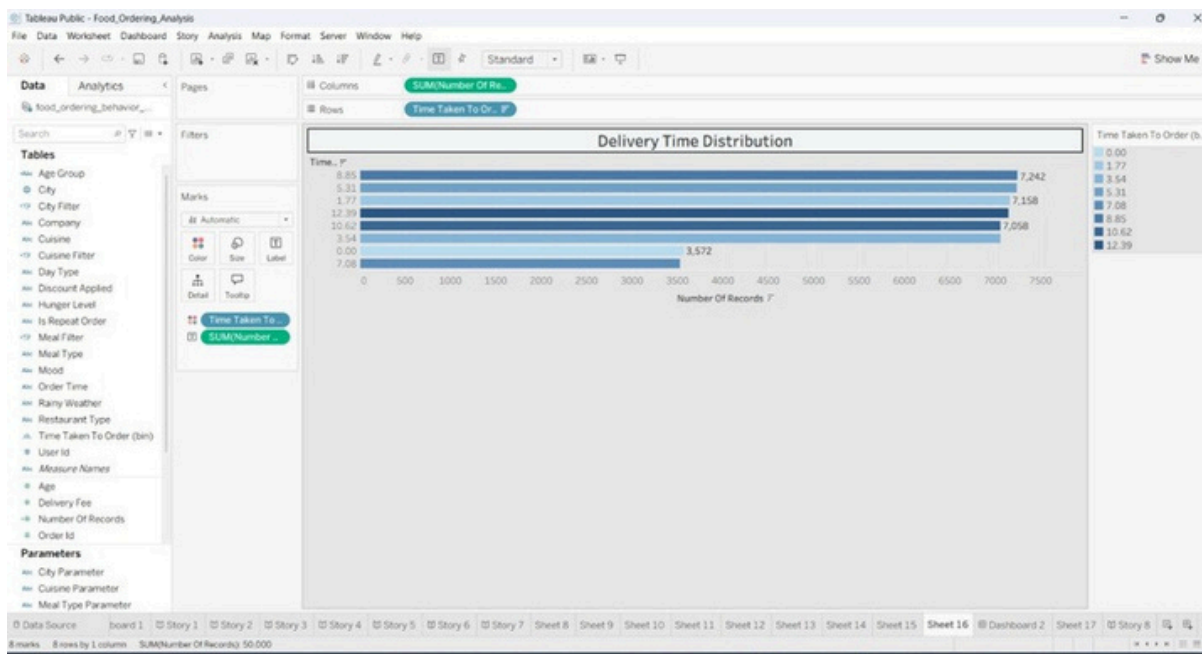
Visualization:



Business Question 7: What is the distribution of delivery time?

Observation: The dashboard shows how delivery times are distributed across all orders.

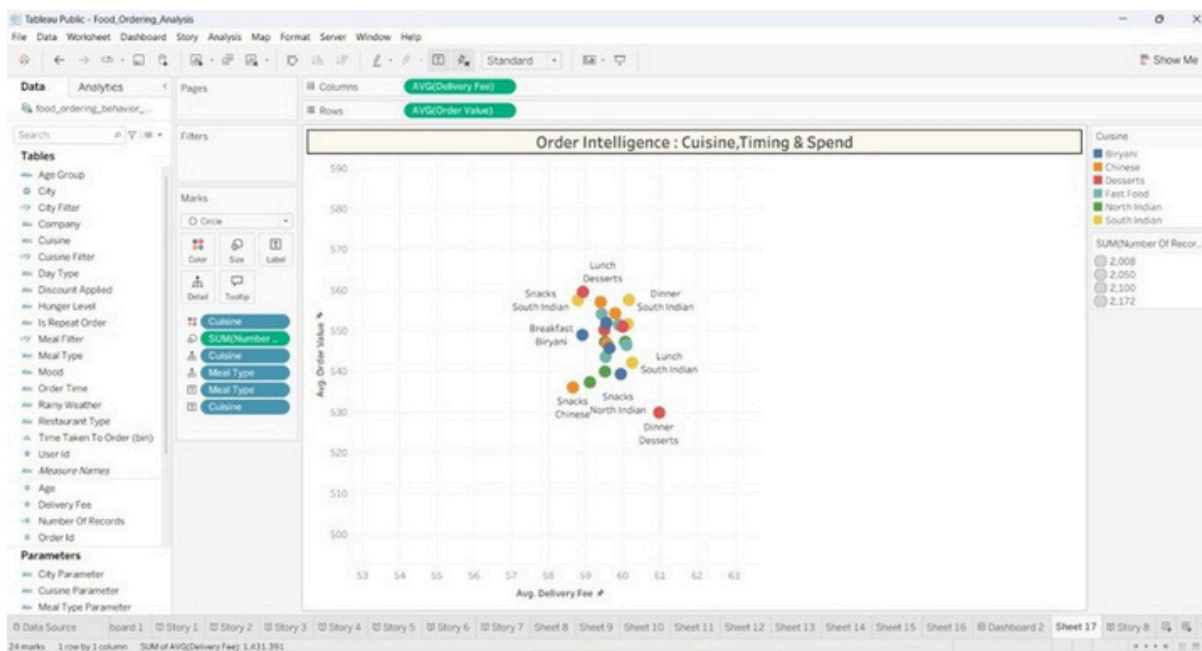
Visualization:



Business Question 8: How do cuisine, order timing and delivery speed influence customer orders?

Observation: The dot chart highlights relationships between cuisine, ordering time and delivery speed.

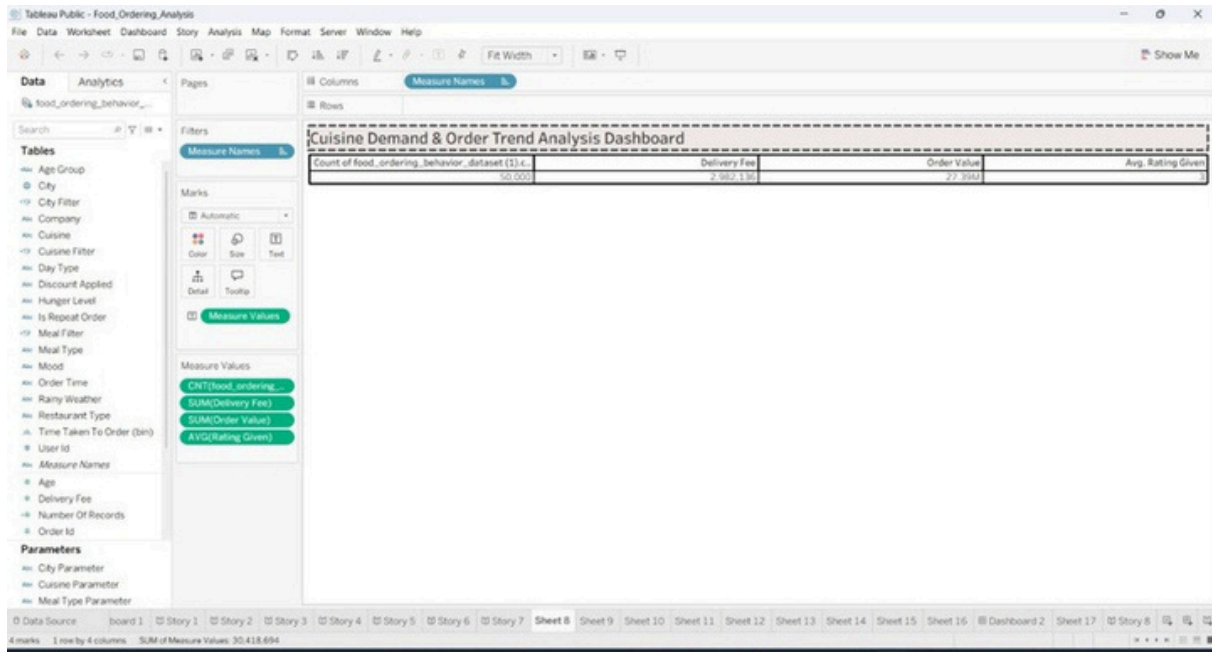
Visualization:



Business Question 9: What are the overall business performance metrics?

Observation: The dashboard displays Total Orders, Total Order Value, Delivery Fee and Average Rating.

Visualization:



Business Question 10: How do cuisine preferences vary across different cities?

Observation: The heatmap shows city-wise cuisine preferences and demand.

Visualization:

